FIG. 14B — PORTABLE CREDENTIAL — NFC ENCODING, VERIFICATION & DELEGATION 1406

SECTION D: NFC TAG NDEF ENCODING 1406

NTAG216 — 500 bytes user memory (pages 4-225, 4 bytes/page)

Page 4 :

```
TLV: 0x03 | 0xFF | hi | lo (NDEF Message, 3-byte length for >255)

NDEF Message 1 — Member ID (Well-Known Text) ~120 bytes (standalone write)
  0xD1 (MB=1,ME=1,SR=1,TNF=0x01) | Type:'T' | Status:0x02 UTF-8 | Lang:"en"
  Payload: walletId string

NDEF Message 2 — Attestation (External Type) ~500 bytes (standalone write)
  0xC4 (MB=1,ME=1,SR=0,TNF=0x04) | Type(26):"zktrident.care:attestation"
  Payload: JSON attestation (Long Record, 4-byte length)

0xFE Terminator TLV Total: ~620 bytes / 888 capacity = 70%
```

T2T Commands: READ(0x30) 16 bytes/page, WRITE(0xA2) 4 bytes/page

NFC HARDWARE TRANSPORT STACK:

NfcBridge (Node.js)	Application — connect(), readTag(), writeTag(), 100ms poll
USB HID Reports (64B)	Transport — node-hid
LpcUsbSioTransport	LPCUSBSIO protocol (I2C over HID), VID:0x1FC9, PID:0x0088
LPC11U24 (USB-to-I2C)	Hardware — MIKROE-2540 Click Board
PN7150 (NCI, I2C@400kHz)	NFC Controller — addr 0x28
NTAG216 (ISO 14443, RF)	Tag — 13.56 MHz, 888 bytes

SECTION E: OFFLINE VERIFICATION FLOWS 1408

PATH A: FULL ZK PROOF (PortableCredential)

Receive via QR / network

↓

Check expiry (expiresAt > now?)

↓

Deserialize: CredentialOperationProof
.fromJSON(proofJson)

↓

Verify: CredentialProgram.verify(proof)
Uses CACHED verification key
No network needed
Pickles recursive verification

↓

RESULT: mathematically proven
TRUST: ZK proof (math certainty)
NEEDS: Cached verification key

Size: 10-22 KB + Transport: QR/network
Verify: ms + Offline: YES
Revocation: expiry only
Proves: ZK credential valid (direct)
Privacy: Highest (ZK)

PATH B: Ed25519 ATTESTATION (NfcCredAttestation)

Patient taps NTAG216 on reader

↓

Parse NDEF: standalone message (walletId or attestation)

↓

Check format: v===2, t==='credential-attestation'

↓

Check expiry: now > att.exp?

↓

Rebuild canonical message, Ed25519.verify(msg,pk,sig)
Time: < 1ms, Network: NONE, pubkey in attestation

↓

(Optional) Online revocation check: isRevoked(att.n)
RESULT: VERIFIED — Ed25519 attestation
TRUST: Cryptographic (Ed25519 signature integrity)

Size: ~500B | Transport: NTAG216 NFC (888B)
Verify: <1ms | Offline: YES
Revocation: expiry + nonce set
Proves: sequencer saw valid ZK proof (receipt)
Privacy: High (only hashes)

SECTION F: DELEGATION CREDENTIAL TTL TIERS 1410

CMS-1696 Appointment of Representative — PortableDelegationCredential:

TTL TIERS:

Scope	TTL	Risk	Description
READ-ONLY	72 hrs	LOW	View records only
WRITE	24 hrs	MED	Modify data
PICKUP	4 hrs	HIGH	Medication dispensing (48h cool-down before use)

Cool-down: 48h before PICKUP activates (prevents instant social engineering)
Rate limit: 24h between new delegations Domain tag: 3 (AOR delegation)

PortableDelegationCredential fields:
version, credentialType:'delegation', delegation: {delegatorWalletId, delegateWalletId,
scopeFlags, effectiveDate, expirationDate, relationshipCode, delegationHash},
delegationIndex, witnessPath[], witnessIsLefts[], delegationRegistryRoot, messagingRoot,
chainStateTimestamp, createdAt